Optimal Monetary Policy according to HANK

Sushant Acharya, Edouard Challe and Keshav Dogra

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Overview

This replication package provides the code to replicate the analysis in our paper, using MATLAB R2020b. One main file runs all the code to generate the data for the figures in the paper and outputs the value of two parameters. The replicator should expect the code to run for about 30 seconds.

Data Availability and Provenance Statements

☒ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

Computational requirements

Software Requirements

Matlab (code was last run with version R2020b). The code was last executed on a 3.5 GHz 6-Core Intel Xeon E5 desktop running Mac OS version 10.15.7.

The following MATLAB toolboxes are required: Optimization Toolbox and Symbolic Math Toolbox. The code was last executed using Optimization Toolbox version 9 and Symbolic Math Toolbox version 8.6.

Memory and Runtime Requirements

The running time of the file is approximately 34 seconds on the desktop machine described above.

Description of programs/code

The programs provided will generate all figures in the main paper and Appendix and will print the value of the parameters delta (δ), Upsilon (Υ) and varkappa ($\varkappa$) for the baseline model presented in Section 4 of the paper. The file `main.m` will run them all. More specifically, `main.m` calls the following scripts:

1. `plot_appendix_figs.m`, which generates Figures H.1, H.2, I.1, J.1 and J.2 in the Online Appendix. `plot_appendix_figs.m` uses (i)

makeec.m, (ii) optw.m, (iii) optdynamics_iid.m, (iv) tcdynamics_iid.m, (v) makeecHTM.m, (vi) optwHTM.m, (vii) optdynamicsHTM.m, (viii) tcdynamicsHTM.m, (ix) makeecPERS.m, (x) optwPERS.m

2. plot_figs_1to3.m, which generates Figures 1, 2 and 3 in the main paper. plot_figs_1to3.m uses (i) realpeg_dynamics_iid.m, (ii) make_compstats.m, (iii) makeec.m and (iv) optw.m

3. plot_figs_4to7.m, which generates Figures 4,5,6, and 7 in the main paper. plot_figs_4to7.m uses (i) makeec.m, (ii) optw.m, (iii) optdynamics_iid.m, (iv) tcdynamics_iid.m and (v) optdynamics_profits.m. plot_figs_4to7.m also prints the values of Upsilon (Υ), delta (δ) and varkappa ($\varkappa$) for the baseline model presented in Section 4.

Instructions to Replicators

Run main.m (without adding a -nodisplay flag) to produce all figures in the paper and Appendix and also to print the values of Upsilon (Υ), delta (δ) and varkappa ($\varkappa$) for the baseline model presented in Section 4. This will save all the figures in *.eps format in the same folder as main.m.

List of tables and programs

There are no tables in the paper. The provided code reproduces:

☒ All figures in the paper and Appendix, as described below
☒ All numbers provided in text in the paper, as described below

Figure	**Program**	**Line Number**	**Output File**
Figure 1	plot_figs_1to3.m	138	figure1.eps
Figure 2	plot_figs_1to3.m	182	figure2.eps
Figure 3	plot_figs_1to3.m	97	figure3.eps
Figure 4	plot_figs_4to7.m	100	figure4.eps
Figure 5	plot_figs_4to7.m	141	figure5.eps
Figure 6	plot_figs_4to7.m	201	figure6.eps
Figure 7	plot_figs_4to7.m	258	figure7.eps
Figure H.1	plot_appendix_figs.m	193	figureH1.eps
Figure H.2	plot_appendix_figs.m	221	figureH2.eps
Figure I.1	plot_appendix_figs.m	282	figureI1.eps
Figure J.1	plot_appendix_figs.m	92	figureJ1.eps
Figure J.2	plot_appendix_figs.m	134	figureJ2.eps

In-text numbers	**Program**	**Line Number**
Upsilon (Υ) for baseline model in Section 4	plot_figs_4to7.m	57
delta (δ) for baseline model in Section 4	plot_figs_4to7.m	58
varkappa ($\varkappa$) for baseline model in Section 4	plot_figs_4to7.m	59